

Vishwash Agarwal

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LinkedIn — GitHub

Professional Summary

Driven fourth-year engineering student pursuing an Integrated MTech in Data Data Science at VIT Bhopal. Focused on developing strong problem-solving skills, mastering web development, and exploring Machine Learning and AI. Enjoys working with mathematics, statistics, and probability. Known for leadership, creativity, and collaboration.

Education

VIT Bhopal University, India

Integrated MTech in Data Science

2022–Present

CGPA: 8.68

St. Joseph's School, Gorakhpur, Uttar Pradesh

Class 12 – 89.4% (2022)

Class 10 – 90.5% (2020)

Skills

Programming: Python, C, C++, JavaScript (ES6)

ML & Deep Learning: MobileNetV2, CNNs, Transfer Learning, Classification, Model Debugging

LLMs & APIs: Gemini API, Prompt Engineering, Chunking Pipelines, Caching, Retry Logic

Web Development: Streamlit, HTML, CSS, ReactJS, NodeJS, ExpressJS

Cloud & Databases: Firebase Auth, Firestore, AWS (EC2, S3, Lambda, RDS)

Data Tools: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, OpenCV

Version Control: Git, GitHub

Problem Solving: Data Structures & Algorithms

Projects

Cheat Day Calorie Tracker (ML + Full Stack)

Technologies: Streamlit, TensorFlow (MobileNetV2), Firebase Auth, Firestore, Python

Live Demo — GitHub

Built a MobileNetV2-based Indian food classifier achieving **97.27% train / 91.75% validation accuracy**. Developed a full-stack Streamlit app with Firebase authentication and Firestore logging. Optimized image preprocessing for faster inference.

Hand Gesture Recognition (Prodigy ML Task)

Technologies: TensorFlow, MobileNetV2, OpenCV, Streamlit, Python

Live App — GitHub

Trained a gesture-recognition CNN on **20,000+ images**, reaching **99.56% training** and **85.8% peak validation accuracy**. Fixed preprocessing mismatch (grayscale→RGB), improving loss from 0.171 to 0.0134. Deployed real-time webcam inference and analyzed domain-shift issues.

YouTube Video Summarizer (LLM Pipeline)

Technologies: Python, Google Gemini API, YouTube Transcript API, Streamlit

GitHub Repository

Built a transcript → chunk → LLM summarization pipeline using Gemini 2.5 Flash with caching, retry logic, and multi-format output. Improved response performance by **40–60%**. Added “Chat with the Video” and automatic export to .txt/.json.

Achievements

- **President, Basketball Club (2020–21):** Led a 25-member team, organized inter-school tournaments, and built strong coordination between players and faculty.
- **Gold Medallist – NASA Olympiad, Science Olympiad, Cyber Olympiad:** Recognized for consistent academic excellence and logical reasoning.
- **Vice-Captain, Green House (2018–19):** Assisted in planning inter-house competitions and improved team participation.
- **Runner-Up – Speech Competition (2019–20):** Awarded for clarity, confidence, and strong communication skills.
- **Community Contributor:** Volunteered in organizing science exhibitions and student mentorship sessions.